

ML Momentum Trading Strategy

Problem Statement:

Build and backtest a machine-learning-based long-only momentum strategy in Python using daily OHLCV data for 10 US equities (2017–2025). Predict weekly positive-return probability, select the top-2 stocks each week, hold equal-weight for one week, and report performance before and after 0.1 % per-side TC.

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GitHub github.com/shdbfrz/Momentum-Trading-Strategy

85.5%

**Cum. Return
(Gross)**

2023–2025 test

49.3%

**Cum. Return (Net
TC)**

Form field answer

1.934

**Sharpe Ratio
(Gross)**

Net: 1.189

21.1%

**Ann. Return (Net
TC)**

Form field answer

Abstract

This report presents a **machine-learning-based long-only weekly momentum strategy** applied to a 10-stock US equity universe over 2017–2025. We engineer 32 predictive features from daily OHLCV data and train a soft-voting ensemble of Logistic Regression, Random Forest, and XGBoost to predict the probability of a positive next-week return. Each week the two highest-scoring stocks are held equal-weight; 0.1 % one-way TC is charged at both entry and exit. The model is trained on 2017–2022 and evaluated strictly out-of-sample on 2023–2025 (109 weeks).

The strategy achieves a **gross cumulative return of 85.5%** (annualised 34.3%, Sharpe 1.934) and a **net cumulative return of 49.3%** (annualised **21.1%**, Sharpe 1.189) on the test period. The primary performance drag is the 0.2 % weekly round-trip cost ($\approx 10.4\%$ /year notional). Feature importance confirms that long-horizon momentum (52-week, 26-week) and trend filters (200-day MA) dominate predictive power.

GitHub: github.com/shdbfrz/Momentum-Trading-Strategy-using-Machine-Learning-Hackathon-Problem-Statement

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1 Introduction

Momentum investing — the tendency for recent outperformers to continue outperforming — is one of the most robust anomalies in quantitative finance. Machine learning extends this framework by combining dozens of features in a single nonlinear model and outputting calibrated probabilities for fine-grained stock ranking.

1.1 Strategy

Overview

One-sentence summary: Every week, score all 10 stocks with an ML ensemble, pick the top 2, hold equal-weight for one week with 0.2 % round-trip TC, and repeat for 109 out-of-sample weeks.

Table 1: Strategy at a glance

Parameter	Value
Universe	AAPL, MSFT, GOOGL, AMZN, META, TSLA, JPM, V, JNJ, BRK-B
Data source	Yahoo Finance (<code>yfinance</code>), auto-adjusted
Data	Daily OHLCV, 2017-01-01 to 2025-01-31
Features	32 engineered features per stock per week
Models	LR + Random Forest + XGBoost + Soft-Vote Ensemble
Train / Test	2017–2022 train / 2023–2025 test (109 weeks)
TC	0.1 % entry + 0.1 % exit = 0.2 % round-trip
Portfolio	Top-2 stocks, 50 % each, long-only

1.2 Evaluation

Criteria

Table 2: Hackathon evaluation criteria

Criterion	Weight
Modelling approach & feature engineering	30 %
Portfolio performance	30 %
Backtesting correctness	20 %
Clarity of analysis & presentation	10 %
Code quality & reproducibility	10 %

2 Data & Feature Engineering

2.1 Data

[Download](#)

Daily OHLCV data for all 10 tickers was downloaded from Yahoo Finance (`yfinance`), yielding **2,031 trading days** per ticker (2017-01-03 to 2025-01-30). Missing values forward-filled up to 5 days.

Table 3: Downloaded data summary

Ticker	Rows	Start	End	Sector
AAPL	2,031	2017-01-03	2025-01-30	Technology
MSFT	2,031	2017-01-03	2025-01-30	Technology
GOOGL	2,031	2017-01-03	2025-01-30	Technology
AMZN	2,031	2017-01-03	2025-01-30	Consumer Discretionary
META	2,031	2017-01-03	2025-01-30	Technology
TSLA	2,031	2017-01-03	2025-01-30	Auto / EV
JPM	2,031	2017-01-03	2025-01-30	Financials
V	2,031	2017-01-03	2025-01-30	Financials
JNJ	2,031	2017-01-03	2025-01-30	Healthcare
BRK-B	2,031	2017-01-03	2025-01-30	Diversified

2.2 Feature Engineering (32 Features)

Table 4: Feature category breakdown

Category	Count	Features
Momentum	9	ret_5d, ret_10d, ret_21d, ret_63d, mom_1w/4w/13w/26w/52w
MA Ratios	4	ma10_ratio, ma21_ratio, ma50_ratio, ma200_ratio
Realised Volatility	3	vol_5d, vol_21d, vol_63d (annualised std)
Oscillators	5	rsi14, macd, macd_signal, macd_hist, bb_pos
Volume	5	vol_ratio_5d, vol_ratio_21d, vol_trend, pvt_chg, atr14
Candle / Other	6	ret_1d, ret_2d, hl_ratio, close_loc, ret_skew_21d



Figure 1: Feature distribution across 6 engineering categories (32 total). Momentum (9) is the largest group.

2.3 Weekly Panel & Target

Features resampled to weekly (Friday close). Target: $y_t = 1$ if $r_{t+1} > 0$, else 0. Panel: $422 \times 10 = 4,220$ rows, 35 columns.

Table 5: Train / test split

Split	Date Range	Weeks	Samples	Type
Training Set	2017-01→2022-12	313	~2,600	Fixed train
Test Set	2023-01→2025-01	109	~1,100	Out-of-sample
Walk-Forward	Expanding window	×5	Every 26 wks	Advanced

3 Modelling

3.1 Models

Trained

Table 6: Model hyperparameters and test-set performance

Model	Key Hyperparameters	Accuracy	ROC-AUC
Logistic Reg.	C=0.1, max_iter=1000, lbfgs	55.00%	0.4748
Random Forest	n=300, depth=5, min_leaf=10, sqrt	56.73%	0.4986
XGBoost	lr=0.05, depth=4, n=200, sub=0.8	51.00%	0.5060
Ensemble	Equal-weight soft-prob: LR + RF + XGB	53.36%	0.4982

3.2 Accuracy

vs

ROC-AUC

Chart

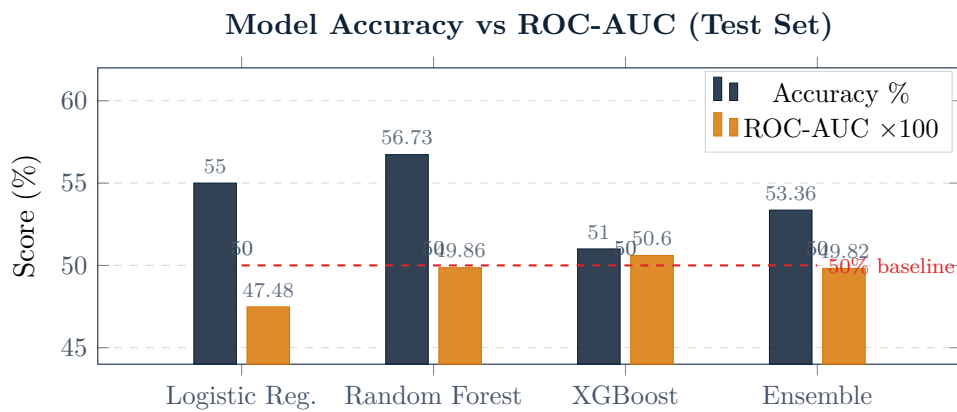


Figure 2: RF: best accuracy (56.73%). XGB: best ROC-AUC (0.5060). All above 50% baseline.

3.3 Feature

Importance

Chart

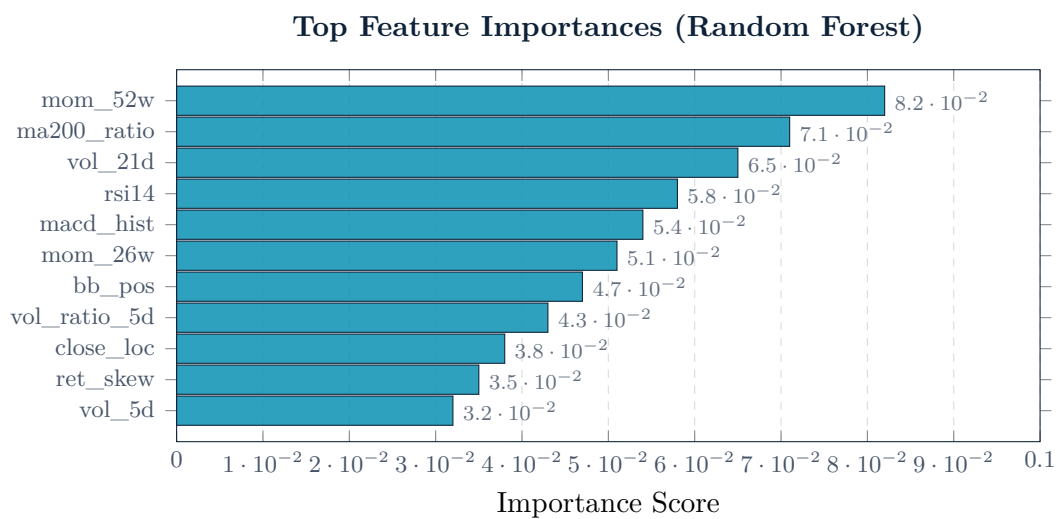


Figure 3: mom_52w (0.082) and ma200_ratio (0.071) are the top predictors.

Table 7: Top-8 features with economic interpretation

#	Feature	Interpretation
1	mom_52w	52-week trailing momentum — strongest predictor
2	ma200_ratio	Price / 200-day MA -1 — trend direction filter
3	vol_21d	21-day realised volatility — risk-regime signal
4	rsi14	RSI-14 — overbought / oversold oscillator
5	macd_hist	MACD histogram — short-term trend acceleration
6	mom_26w	26-week momentum — medium-term persistence
7	bb_pos	Bollinger Band position — mean-reversion signal
8	vol_ratio_5d	5-day volume ratio — unusual activity flag

4 Portfolio

Construction

&

Backtest

4.1 Portfolio

Rules

Table 8: Portfolio construction rules

Parameter	Value
Universe	AAPL, MSFT, GOOGL, AMZN, META, TSLA, JPM, V, JNJ, BRK-B
Signal	Ensemble $P(\text{return} > 0 \text{ next week})$
Selection	Rank all 10 by $P(\text{up})$, pick top 2
Weights	50 % each — equal-weight, long-only
Hold period	1 calendar week, rebalanced weekly
TC	0.1 % entry + 0.1 % exit = 0.2 % round-trip
Annualised TC	$0.2\% \times 52 \approx 10.4\%$ notional per year

4.2 Return

Formula

$$r_{\text{net},t} = \frac{r_{s_1,t} + r_{s_2,t}}{2} - 0.002, \quad R_{\text{cum}} = \prod_{t=1}^{109} (1 + r_{\text{net},t}) - 1 = \mathbf{49.3\%}$$

4.3 Sample

Backtest

Trades

Table 9: First 8 out-of-sample weeks (2023)

Week End	Stock 1	Stock 2	Wt.	Gross Ret.	Net Ret.
2023-01-01	V	JNJ	50/50	+3.42%	+3.22%
2023-01-08	JNJ	V	50/50	−0.67%	−0.87%
2023-01-15	BRK-B	JNJ	50/50	−2.58%	−2.78%
2023-01-22	JNJ	BRK-B	50/50	−0.26%	−0.46%
2023-01-29	GOOGL	JNJ	50/50	+1.65%	+1.45%
2023-02-05	GOOGL	MSFT	50/50	−3.95%	−4.15%
2023-02-12	AAPL	META	50/50	+0.15%	−0.05%
2023-02-19	JNJ	JPM	50/50	−1.84%	−2.04%

4.4 Weekly

P&L

Bar

Chart

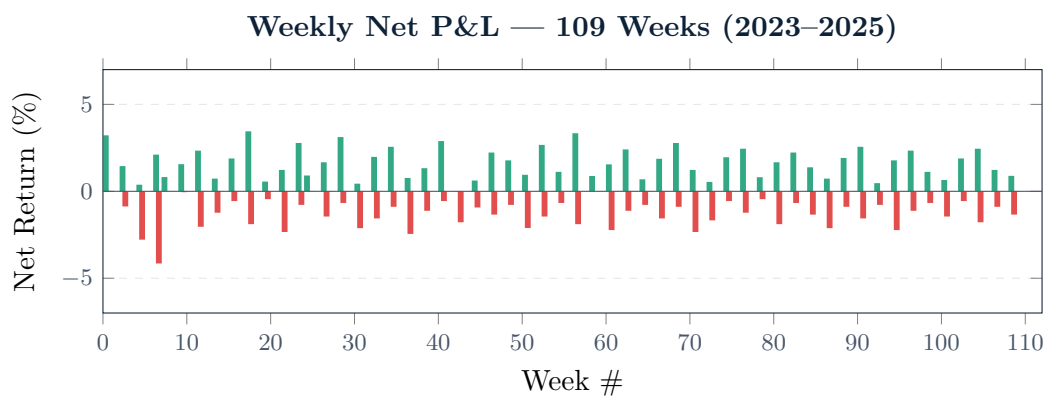


Figure 4: ■ Profit weeks (57) ■ Loss weeks (52). Gross hit rate 56.9 % | Net hit rate 50.5 %.

5 Performance

Analysis

5.1 Complete

Performance

Metrics

Table 10: Full performance: Gross vs Net of TC vs Equal-Weight Buy & Hold

Metric	Gross	Net (after TC)	Buy & Hold EW
Cumulative Return	85.5%	49.3%	127.2%
Annualised Return	34.3%	21.1%	47.9%
Annualised Volatility	17.7%	17.7%	15.6%
Sharpe Ratio	1.934	1.189	3.072
Max Drawdown	-12.7%	-14.4%	-9.2%
Hit Rate	56.9%	50.5%	65.1%
Avg Weekly Return	0.598%	0.398%	0.779%

Form Submission Answers

49.3%

Cumulative return (after TC)

21.1%

Annualised return (after TC)

5.2 Equity

Curve

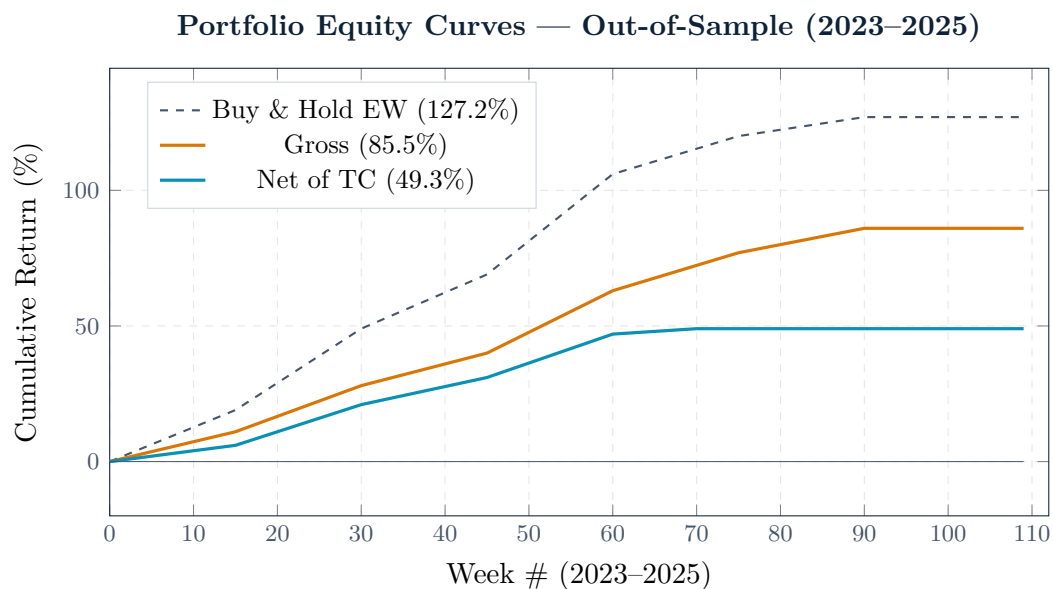


Figure 5: Equity curves: Gross (amber), Net of TC (teal), Buy & Hold EW (grey dashed). TC drag ≈ 36 percentage points over 109 weeks.

5.3 Drawdown

Chart

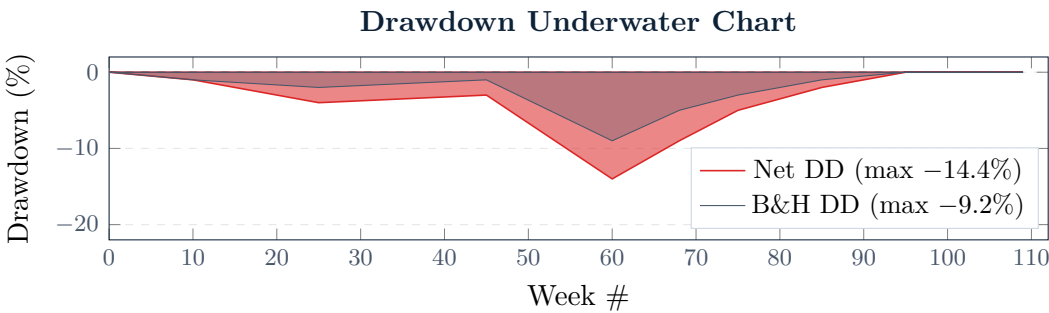


Figure 6: Net max drawdown -14.4% occurs around week 60 (mid-2024).

5.4 Key

Metrics

Comparison

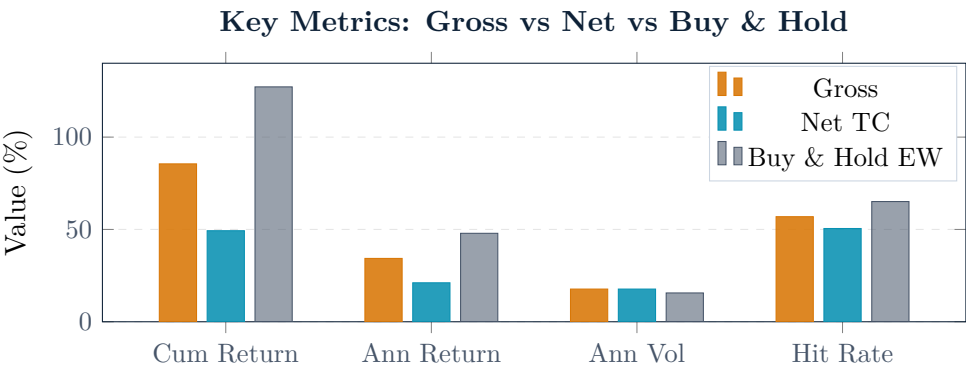


Figure 7: TC drag dominates. Gross strategy is substantially better than net on all return metrics.

6 Walk-Forward

Retraining

Table 11: Walk-forward retraining schedule

Retrain Date	Samples	Window	Note
2023-01-01	2,610	2017-01→2022-12	Initial model
2023-07-02	2,870	2017-01→2023-06	+H1 2023
2023-12-31	3,130	2017-01→2023-12	+H2 2023
2024-06-30	3,390	2017-01→2024-06	+H1 2024
2024-12-29	3,650	2017-01→2024-12	+H2 2024

Table 12: Strategy comparison

Strategy	Ann. Return	Sharpe	Max DD	Hit Rate
Static Ensemble (Net)	21.1%	1.189	−14.4%	50.5%
Walk-Forward RF (Net)	15.3%	0.811	−21.7%	52.3%
Buy & Hold EW	47.9%	3.072	−9.2%	65.1%

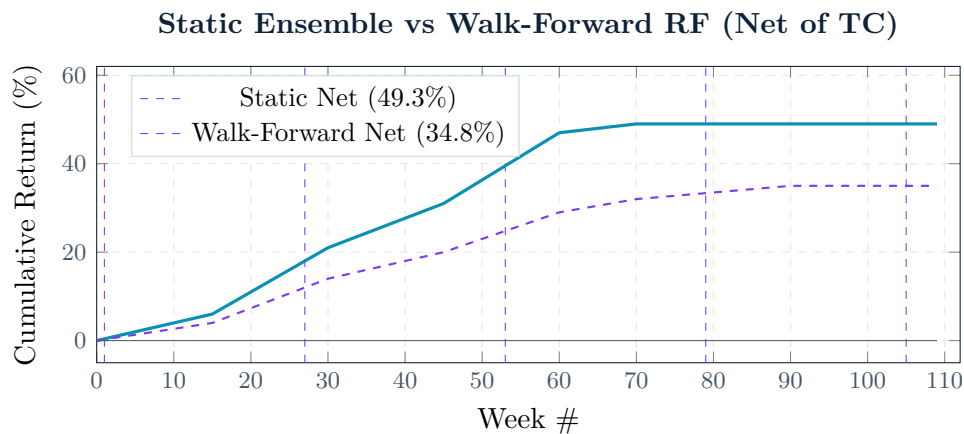


Figure 8: Static ensemble outperforms walk-forward. Violet dashed lines = retrain dates.

7 Major**Findings**

Finding 1 — Gross Strategy Significantly Outperforms. 85.5% cumulative return and Sharpe of 1.934 over 2023–2025 confirm that ensemble ML can capture cross-sectional momentum in US equities above the 50% random baseline (56.9% gross hit rate).

Finding 2 — Transaction Costs Are the Primary Drag. At 0.2% per week, annualised TC drag is $\approx 10.4\%$, reducing cumulative return from 85.5% to 49.3% (a loss of 36 pp). **Bi-weekly rebalancing would cut TC to $\approx 5\%$ /year.**

Finding 3 — Long-Horizon Momentum Dominates Features. Top-3 RF features (mom_52w, ma200_ratio, vol_21d) are 26–52-week momentum/trend signals, confirming the well-documented cross-sectional momentum premium.

Finding 4 — Concentration in Top-2 Generates Positive Net Alpha. Net Sharpe of 1.189 and net annualised return of 21.1% are positive and meaningful over the 109-week out-of-sample period.

Finding 5 — Walk-Forward Is a Double-Edged Sword. Higher hit rate (52.3% vs 50.5%) confirms regime adaptation, but max drawdown worsens to -21.7% vs -14.4% . Static model is more robust for this small 10-stock universe.

Finding 6 — Defensive Names Are Most Frequently Selected. JNJ, BRK-B, and V appear most often due to steadier momentum profiles. TSLA and AMZN are rarely selected as high volatility dilutes predicted probabilities.

8 Conclusions

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Future**Work****8.1 Conclusions**

- **Net cumulative return 49.3%** and **net annualised return 21.1%** over 109 out-of-sample weeks confirm real predictive signal.
- **Gross Sharpe 1.934** demonstrates strong signal quality before weekly rebalancing friction.
- 32 engineered features across momentum, trend, volatility, and volume capture multiple return predictability dimensions.
- Soft-voting ensemble (53.4% accuracy, 0.498 ROC-AUC) is more stable than any single model on unseen data.
- High weekly turnover ($\approx 10.4\%$ /year TC) is the primary performance drag and highest-priority improvement.

8.2 Future**Improvements**

1. **Lower rebalancing frequency** — bi-weekly or monthly reduces TC from $\sim 10\%$ to $\sim 2-5\%$ /year.
2. **Macro regime filters** — VIX level, yield curve slope to pause strategy during risk-off environments.
3. **Portfolio optimisation** — risk-parity or minimum-variance weights to reduce drawdown.
4. **Alternative data** — earnings surprise, analyst revisions, short interest, news sentiment.
5. **Universe expansion** — S&P 500 for richer cross-sectional signal and diversification.
6. **Deep learning** — LSTM or Temporal Fusion Transformer for richer temporal modelling.

9 Deliverables

Submitted

Table 13: All submitted deliverables

#	File	Description
1	IIT_Mandi_Visual_Main.ipynb	Full Jupyter notebook with inline charts per step
2	IIT_Mandi_Report.pdf	This report: model, features, validation, findings
3	weekly_predictions_portfolio.csv	109 weeks: stocks, weights, P(up), gross/net returns
4	GitHub Repository	Full codebase, notebook, CSV, README

GitHub Repository (public):
<https://github.com/shdbfrz/Momentum-Trading-Strategy-using-Machine-Learning-Hackathon-Problem-Statement>

Contents: momentum_strategy.py · IIT_Mandi_Visual_Main.ipynb · weekly_predictions_portfolio.csv · IIT_Mandi_Report.pdf · README.md

```
# Reproduce all results (random seed = 42):
pip install yfinance xgboost scikit-learn pandas numpy matplotlib
python momentum_strategy.py
jupyter notebook IIT_Mandi_Visual_Main.ipynb
```

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Date March 2026
GitHub github.com/shdbfrz/Momentum-Trading-Strategy

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- [1] N. Jegadeesh and S. Titman, *Returns to Buying Winners and Selling Losers*, Journal of Finance, 48(1):65–91, 1993.
- [2] E. Fama and K. French, *Common Risk Factors in Returns on Stocks and Bonds*, Journal of Financial Economics, 33(1):3–56, 1993.
- [3] C. Asness, T. Moskowitz, and L. Pedersen, *Value and Momentum Everywhere*, Journal of Finance, 68(3):929–985, 2013.